

Name:	_____
Class:	12MTZ1
Teacher:	MR KNOX

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2015 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 2

*Time allowed - 3 HOURS
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- Multiple choice questions are to be answered on the multiple choice sheet provided at the back of the paper.
- Each question in the extended response is to be commenced in a new booklet clearly marked Question 11, Question 12, etc on the top of the page and must show your name and class.
- All necessary working should be shown in every question for extended response. Full marks may not be awarded for careless or badly arranged work.
- Board of Studies approved calculators may be used. Standard Integral Tables are provided.
- Write your name and class in the space provided at the top of this question paper.

1) Which pair of equations gives the directrices of $y = \frac{4}{x}$

- (A) $y = \pm(x + 2\sqrt{2})$
- (B) $y = -x \pm 4\sqrt{2}$
- (C) $y = \pm(x + 4\sqrt{2})$
- (D) $y = -x \pm 2\sqrt{2}$

2) Which expression is equal to $\int \sec x dx$

- (A) $\sec x \tan x + C$
- (B) $\tan x + C$
- (C) $\ln|\sec x + \tan x| + C$
- (D) $\ln|\sin x| + C$

3) For a complex number $x + yi = \sqrt{2} + 3i$

- (A) $x^2 + y^2 = 3$ and $2xy = 2$
- (B) $x^2 + y^2 = 2$ and $-2xy = 3$
- (C) $x^2 - y^2 = 2$ and $2xy = 3$
- (D) $x^2 + y^2 = 2$ and $2xy = 3$

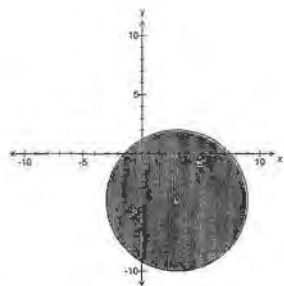
4) For the relation $x^2 + e^y = 7$

- (A) $\frac{dy}{dx}(2x + e^y) = 0$
- (B) $\frac{dy}{dx} = -\frac{2x}{e^y}$
- (C) $\frac{dy}{dx} = 2x$
- (D) $\frac{dy}{dx} 2x + \frac{dx}{dy} e^y = 0$

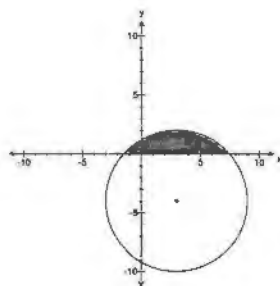
- 5) Which region on the argand diagram shows the complex number z , which is defined by $|z - 3 + 4i| \leq 6$ and $\text{Im}(z) \geq 0$?

The centre of each circle is shown for convenience

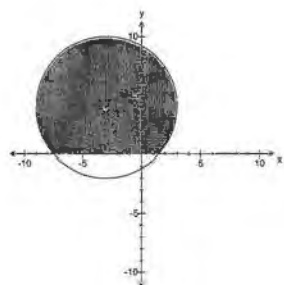
(A)



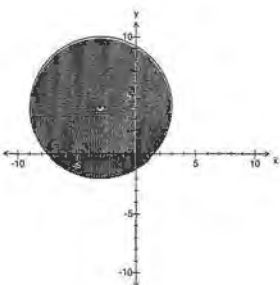
(B)



(C)



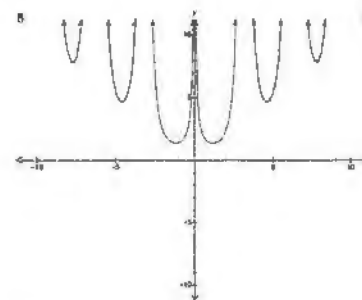
(D)



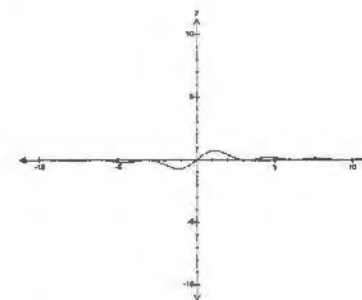
- 6) Which diagram best represents the graph

$$y = \frac{x}{\sin^2 x}$$

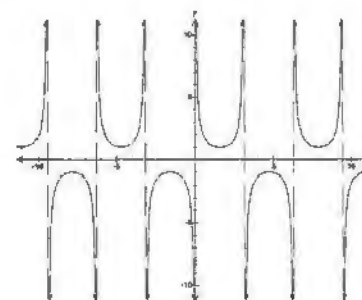
(A)



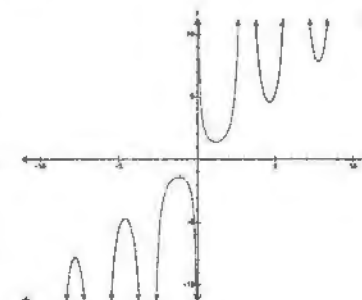
(B)



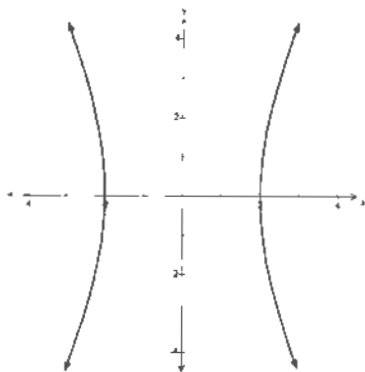
(C)



(D)



Questions 7 and 8 refer to the graph of $\frac{x^2}{4} - \frac{y^2}{16} = 1$



7) What is the eccentricity?

- (A) $\sqrt{2}$
- (B) $\sqrt{\frac{3}{2}}$
- (C) $\frac{\sqrt{5}}{2}$
- (D) $\sqrt{5}$

8) What are the equations of the asymptotes?

- (A) $y = \pm 2x$
- (B) $y = \pm \frac{1}{2}x$
- (C) $y = \pm 2$
- (D) $y = \pm \frac{1}{2}$

9) The equation $x^4 + px + q = 0$, where $p \neq 0$ and $q \neq 0$, has roots α, β, γ and δ .

What is the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$?

- (A) $-4q$
- (B) $p^2 - 2q$
- (C) $p^4 - 2q$
- (D) p^4

10) What is the value of $S = \sum_{r=1}^{\infty} rp(1-p)^{r-1}$?

- (A) $S = 1$
- (B) $S = \frac{1}{p}$
- (C) $S = \frac{1}{1-p}$
- (D) $S = \frac{1}{p(1-p)}$

Question 11 (15 Marks)

- a) Let $z = 1 + i\sqrt{3}$ and $w = -\sqrt{3} + i$
- (i) Express w in terms of z . 1
- (ii) Express z in modulus-argument form. 2
- (iii) Write z^{10} in the form $a + ib$ 2
- b) Find numbers A , B and C such that
- $$\frac{6x^2 + 9x + 4}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$
- 2
- c) Find the value of z if $z^2 + 3iz = 7$ 2
- d) Evaluate $\int_0^{1/2} \frac{x}{\sqrt{1-x^2}} dx$ 3
- e) Sketch on the argand diagram the region defined by
- $$\frac{\pi}{4} < \operatorname{Re}(z) < \frac{\pi}{2} \quad \text{AND} \quad \frac{\pi}{4} < \operatorname{Im}(z) < \frac{\pi}{2} \quad \text{AND} \quad \frac{\pi}{4} < \arg(z) < \frac{\pi}{2}$$
- 3

Question 12 (15 Marks)

- a) (i) Show that for $t = \tan \frac{x}{2}$, 1
- $$dx = \frac{2dt}{1+t^2}$$
- (ii) Use the substitution $t = \tan \frac{x}{2}$, to evaluate 3
- $$\int_0^{\pi/2} \frac{1}{1 + \cos x + \sin x} dx$$
- b) By using an appropriate substitution, evaluate $\int_3^6 \frac{x}{(x+1)\sqrt{x+1}} dx$ 4
- Hint: Consider a substitution which removes the $\sqrt{\quad}$*
- c) Sketch the graph of $y = \frac{x^3 + 3x^2 + x + 4}{x+3}$, showing all asymptotes 3
- d) Find and sketch the locus of the complex number z , where 4
- $$3|z - (2 + 2i)| = |z - (6 + 6i)|$$

Question 13 (15 Marks)

- (a) i) Given $I_n = \int x(x^5 + 1)^n dx$, show that

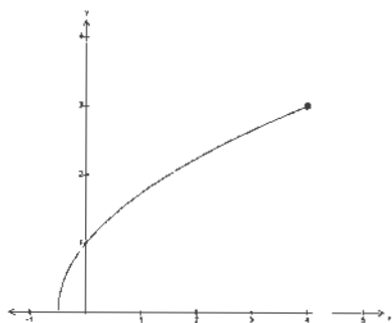
3

$$I_n = \frac{x^2(x^5 + 1)^n}{2 + 5n} + \frac{5n}{2 + 5n} I_{n-1}$$

- ii) Hence, find I_{10} in terms of I_8 and x .

2

- (b) The following graph of $y = f(x)$ starts at x intercept $= -\frac{1}{2}$ and ends at the point $(4,3)$. Sketch



(i) $y = \frac{1}{f(x)}$

1

(ii) $y = f(x - 1) + 1$

1

(iii) $y = \log(f(x))$

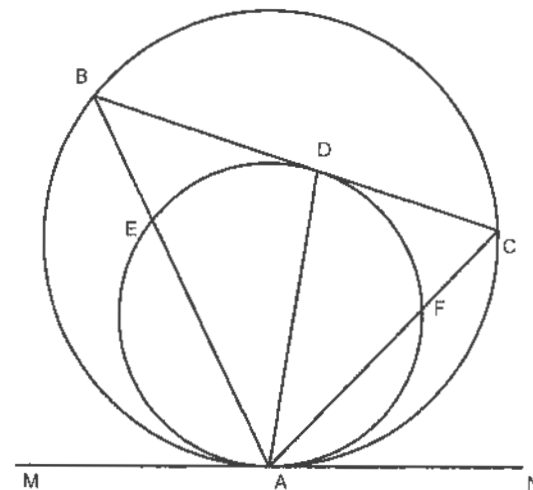
2

(iv) $y = f(x^2)$

2

- (c) In the diagram, MAN is the common tangent of two circles touching internally at A . B and C are two points on the larger circle such that BC is a tangent to the smaller circle with point of contact D . AB and AC cut the smaller circle at E and F respectively. Show that AD bisects $\angle BAC$.

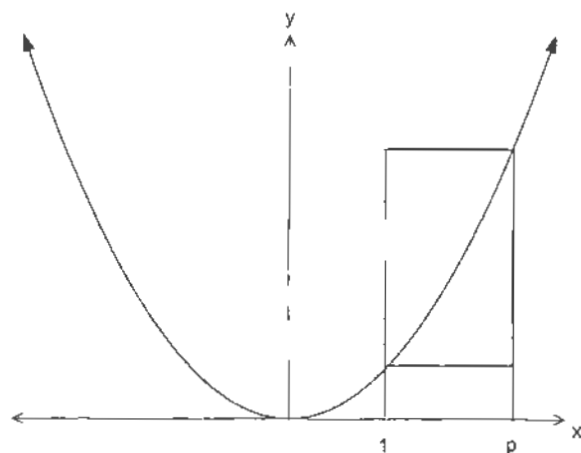
4



Question 14 (15 Marks)

- a) Copy the graph of $y = x^2$ into your writing booklet.

3



By considering this graph, show that $p - 1 < \frac{p^3}{3} - \frac{1}{3} < p^3 - p^2$ for $p > 1$

- b) In a series $T_1 = 3, T_2 = 8$ and $T_n = 2T_{n-1} - T_{n-2}$ for $n > 2$. Show that $T_n = 5n - 2$ for $n \geq 1$.

3

- c) The polynomial $P(x) = x^6 + ax^3 + bx^2$ has a factor $(x + 1)^2$. Find the value of the real numbers a and b .

3

- d) The equation $x^4 + bx^3 + cx^2 + dx + k = 0$ has roots $\alpha, \frac{1}{\alpha}, \beta$ and $\frac{1}{\beta}$.

- (i) Show that $k = 1$

1

- (ii) Show $b = d$

2

- e) The equation $x^3 + 3x^2 + 2x + 1 = 0$ has roots α, β and γ . Find the monic cubic equation with roots α^2, β^2 and γ^2 .

3

Question 15 (15 Marks)

- a) A rectangular hyperbola has the equation $xy = \frac{1}{2}a^2$.

- (i) Show that the point $P\left(\frac{at}{2}, \frac{a}{t}\right)$ lies on the hyperbola, for all values of t .

1

- (ii) Show that the equation of the tangent at P is given by the equation $2x + t^2y = 2at$.

2

- (iii) One of the foci of the hyperbola is $S(a, a)$. DO NOT PROVE THIS. From a point T on the tangent a line is drawn through S , such that PS is perpendicular to PT . Show that the equation of PS is

$$t^2x - 2y = at^2 - 2a$$

- (iv) Hence find the locus of the point T .

2

- b) (i) By starting with $\left(\cos \frac{x}{2} + i \sin \frac{x}{2}\right)^2$ and using De Moivre's Theorem find expressions for $\sin x$ and $\cos x$ in terms of $\sin \frac{x}{2}$ and $\cos \frac{x}{2}$.

2

- (ii) By using the results from part (i) and $t = \tan \frac{x}{2}$ prove the result

1

$$\tan x = \frac{2t}{1-t^2}$$

- c) Let $z = \cos \theta + i \sin \theta$ and let $\omega = \cos \alpha + i \sin \alpha$

- (i) Solve $z^7 = 1$

1

- (ii) Show that $\cos n\alpha = \frac{1}{2}(\omega^n + \omega^{-n})$

1

- (iii) Suppose that $\frac{1}{2} + \cos \alpha + \cos 2\alpha + \cos 3\alpha = 0$. Use part (ii) to show that $\omega^7 = 1$.

2

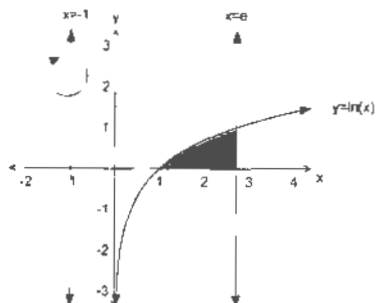
- (iv) Hence solve $\frac{1}{2} + \cos \alpha + \cos 2\alpha + \cos 3\alpha = 0$ for $\pi < \alpha \leq \pi$

1

Question 16 (15 Marks)

- a) The diagram shows the area bounded by the function $y = \ln(x)$, the x-axis and the line $x = e$.

4



Find the volume formed when the area is rotated around the line $x = -1$.

- b) A 20 kg trolley is pushed with a force of 100 N . Friction causes a resistive force which is proportional to the square of the trolley's velocity.

- (i) Show that $\ddot{x} = 5 - \frac{kV^2}{20}$, where k is a positive constant.

1

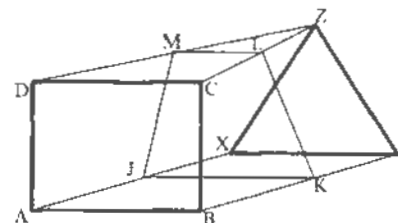
- (ii) If the trolley is initially stationary at the origin, show that the distance travelled when its speed is V is given by

4

$$x = \frac{10}{k} \ln\left(\frac{100}{100 - kV^2}\right)$$

QUESTION 16 CONTINUES NEXT PAGE

- c) In the diagram below, a solid is shown which has a rectangular front which is parallel to the triangular back and the distance between the front and back is $30\pi\text{ cm}$.



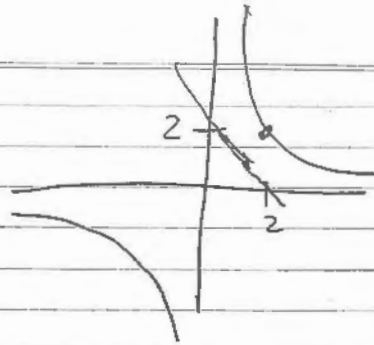
Also $AB = DC = XY = 20\text{ cm}$, $DA = 8\text{ cm}$ and the perpendicular distance from Z to XY is also 8 cm .

- (i) Copy the diagram into your writing booklet and add suitable coordinate axes to your diagram. 1
- (ii) Find expressions for the length of JK and ML . 2
- (iii) Find the area of the cross section $JKLM$. 1
- (iv) Hence find the volume of the solid. 2

END OF PAPER

①

$$y = \frac{4}{x}$$



$$e = \sqrt{2}$$

directrix passes through $\left(\frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}\right)$ ⑤
 $= (\sqrt{2}, \sqrt{2})$

$$m = -1$$

$$\begin{aligned} y - \sqrt{2} &= -1(x - \sqrt{2}) \\ y &= -x + \sqrt{2} + \sqrt{2} \\ y &= -x + 2\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{other is } y + \sqrt{2} &= -1(x + \sqrt{2}) \\ y &= -x - 2\sqrt{2} \end{aligned}$$

Incorrect ① - places $\pm$ at front.

Incorrect ② - Wrong point on directrix used.
 i.e. $(2\sqrt{2}, 2\sqrt{2})$

$$\begin{aligned} y - 2\sqrt{2} &= -(x - 2\sqrt{2}) \\ y &= -x + 4\sqrt{2} \end{aligned}$$

Incorrect ③ $y = \pm(x + 4\sqrt{2})$

error ④ and ⑤

② ③ Correct.

$$\int \sec x \, dx = \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

$$\left[\begin{aligned} \text{let } u &= \sec x + \tan x \\ du &= (\sec x \tan x + \sec^2 x) \, dx \end{aligned} \right]$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

$$= \int \frac{1}{u} \, du$$

$$= \ln |u|$$

$$= \ln |\sec x + \tan x|$$

④ ⑤ incorrect std Integral.

⑥ incorrect

$$\text{Wrong } \int \sec x \, dx = \int (\cos x)^{-1} \, dx$$

$$= \ln(\sin x) \text{ Wrong.}$$

$$\begin{aligned}
 (3) \quad x+iy &= \sqrt{2+3i} \\
 (x+iy)^2 &= 2+3i \\
 x^2+2ixy-y^2 &= 2+3i \\
 x^2-y^2=2 \quad 2xy=3 & \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad x^2+e^y &= 7 \\
 2x + \frac{dy}{dx} e^y &= 0 \\
 \frac{dy}{dx} e^y &= 0 - 2x \\
 \frac{dy}{dx} &= \frac{-2x}{e^y} \quad (B)
 \end{aligned}$$

$$\begin{aligned}
 (5) \quad \text{Circle is centred at } (3, -4) \\
 \text{radius of } 6 \\
 \text{Im}(z) > 0 \rightarrow \text{above } x\text{-axis.} \quad (B)
 \end{aligned}$$

$$\begin{aligned}
 (6) \quad \frac{\text{Top}}{\text{Bottom}} = \frac{-}{+} \quad \frac{\text{Top}}{\text{Bottom}} = \frac{+}{+} = 1 \\
 \text{undefined when } \sin x = 0 \text{ i.e. } x = 0, \pi, 2\pi \\
 (D)
 \end{aligned}$$

$$\begin{aligned}
 (7) \quad b^2 &= a^2(e^2 - 1) \\
 16 &= 4(e^2 - 1) \\
 4 &= e^2 - 1 \\
 5 &= e^2 \\
 e &= \sqrt{5} \quad (D)
 \end{aligned}$$

$$(8) \quad y = \pm \frac{b}{a} x \rightarrow y = \pm 2x \quad (A)$$

$$\begin{aligned}
 (9) \quad \left. \begin{aligned} \alpha^4 + p\alpha + q &= 0 \\ \beta^4 + p\beta + q &= 0 \\ \gamma^4 + p\gamma + q &= 0 \\ \delta^4 + p\delta + q &= 0 \end{aligned} \right\} \begin{aligned} \alpha^4 + \beta^4 + \gamma^4 + \delta^4 &= -p(\alpha + \beta + \gamma + \delta) - 4q \\ \text{by } \alpha + \beta + \gamma + \delta &= -\frac{b}{a} = 0 \\ \alpha^4 + \beta^4 + \gamma^4 + \delta^4 &= -4q \end{aligned} \quad (A)
 \end{aligned}$$

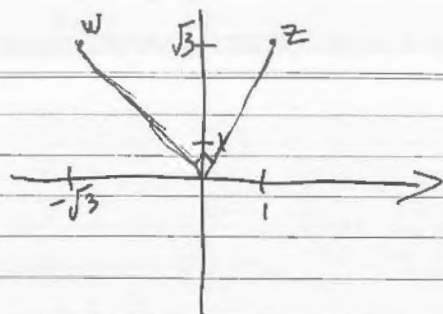
$$(10) \quad S = p + 2p(1-p) + 3p(1-p)^2 + \dots \quad (1)$$

$$(1-p)S = p(1-p) + 2p(1-p)^2 + 3p(1-p)^3 + \dots \quad (2)$$

$$\begin{aligned}
 (1) - (2) \\
 pS &= p + p(1-p) + p(1-p)^2 + \dots \\
 S &= 1 + (1-p) + (1-p)^2 + \dots \\
 a &= 1 \quad r = 1-p
 \end{aligned}$$

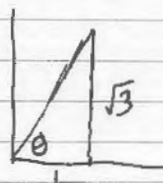
$$\begin{aligned}
 S_{\infty} &= \frac{1}{1-(1-p)} \\
 &= \frac{1}{p} \quad (B)
 \end{aligned}$$

(11) (a) (i)



$$w = z i$$

(ii)



$$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\tan(\arg z) = \frac{\sqrt{3}}{1}$$

$$\arg z = \pi/3$$

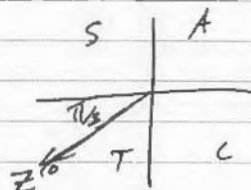
$$z = 2(\cos \pi/3 + i \sin \pi/3)$$

$$(iii) z^{10} = 2^{10} (\cos \pi/3 + i \sin \pi/3)^{10}$$

$$= 1024 (\cos \frac{\pi}{3} \times 10 + i \sin \frac{\pi}{3} \times 10)$$

$$= -1024 \times \frac{1}{2} + i (-1024 \times \frac{\sqrt{3}}{2})$$

$$= -512 - 512\sqrt{3}i$$



11

$$(b) \frac{6x^2 + 9x + 4}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

$$6x^2 + 9x + 4 = A(x^2 + 1) + (Bx + C)x$$

$$x^2: 6 = A + B \quad (1)$$

$$x: 9 = C \quad (2)$$

$$\text{const: } A = 1 \quad (3)$$

from (1) & (3)

$$B = 5$$

$$A = 1, B = 5, C = 9$$

$$(c) z^2 + 3iz = 7$$

$$z^2 + 3iz - 7 = 0$$

$$z = \frac{-3i \pm \sqrt{-9 + 28}}{2}$$

$$= \frac{-3i \pm \sqrt{19}}{2}$$

$$= \pm \frac{\sqrt{19}}{2} - \frac{3i}{2}$$

(d)

$$\textcircled{a} \int_0^{\pi/2} \frac{x}{\sqrt{1-x^2}} dx$$

$$\text{let } u = 1 - x^2$$

$$du = -2x dx \rightarrow x dx = \frac{du}{-2}$$

$$\text{When } x = 0$$

$$u = 1 - 0^2 = 1$$

$$\text{when } x = \pi/2$$

$$u = 1 - (\pi/2)^2 = 1 - \pi^2/4$$

$$= \int_1^{1-\pi^2/4} \frac{1}{\sqrt{u}} \frac{du}{-2}$$

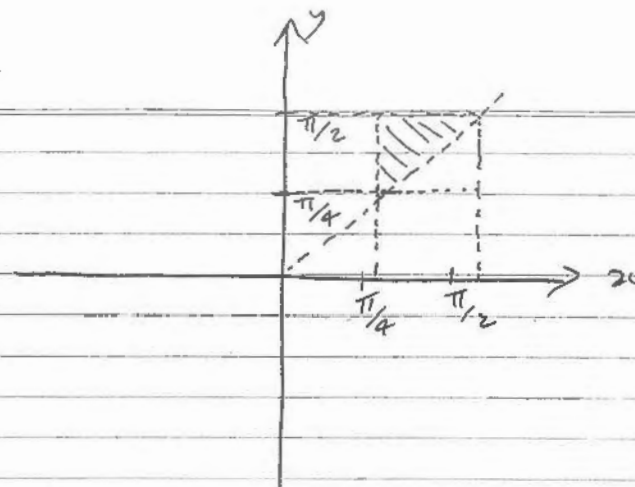
$$= \frac{1}{-2} \int_1^{1-\pi^2/4} u^{-1/2} du$$

$$= \left[2u^{1/2} \right]_1^{1-\pi^2/4}$$

$$= 2\sqrt{1-\pi^2/4} - 2$$

$$= \sqrt{3} - 2$$

$\textcircled{b}$



⑫

②(1) $t = \tan \frac{x}{2}$

$\tan^{-1} t = \frac{x}{2}$

$2 \tan^{-1} t = x$

$\frac{dx}{dt} = 2 \frac{1}{1+t^2}$

$dx = \frac{2dt}{1+t^2}$

(11)

when $x = \pi/2$
 $t = \tan \frac{\pi}{4}$

$= 1$

$x = 0$

$t = \tan 0$

$= 0$

$= \int_0^1 \frac{1}{\frac{1-t^2}{1+t} + \frac{2t}{1+t^2}} \frac{2dt}{1+t^2}$

$= \int_0^1 \frac{1}{\frac{t^2+2t}{1+t^2}} \frac{2dt}{1+t^2}$

$= 2 \int_0^1 \frac{1}{1+2t-t^2} dt$

$=$

⑫ ② (11)

$\int_0^{\pi/2} \frac{1}{1+\cos x + \sin x} dx$

$x = \pi/2$ $t = \tan \frac{\pi}{4}$

$= 1$

$x = 0$

$t = \tan 0$

$= 0$

$= \int_0^1 \frac{1}{\frac{1+2t}{1+t^2} + \frac{1+t^2}{1+t^2}} \frac{2dt}{1+t^2}$ ✓

$= 2 \int_0^1 \frac{2(1+t^2)}{1+t^2+2t+1-t^2} dt$

$= \int_0^1 \frac{1}{t+1} dt$ ✓

$= [\ln(t+1)]_0^1$

$= \ln 2 - \ln 1$

$= \ln 2$ ✓

12) b)

$$\int_3^8 \frac{x}{(x+1)\sqrt{x+1}} dx$$

$$\left. \begin{array}{l} \text{let } u^2 = x+1 \\ x = u^2 - 1 \\ dx = 2u du \end{array} \right\} \checkmark$$

$$\left. \begin{array}{ll} \text{when } x=3 & x=8 \\ u^2 = 3+1 & u^2 = 8+1 \\ u = 2 & u = 3 \end{array} \right\} \checkmark$$

$$= \int_2^3 \frac{u^2 - 1}{u^2 - u} 2u du$$

$$= \int_2^3 \frac{(u+1)(u-1)}{u(u-1)} 2u du \checkmark$$

$$= \int_2^3 2(u+1) du \quad \checkmark \text{ full here}$$

$$= 2 \left[\frac{u^2}{2} + u \right]_2^3$$

$$= 2 \left[\frac{9}{2} + 3 - 2 - 2 \right]$$

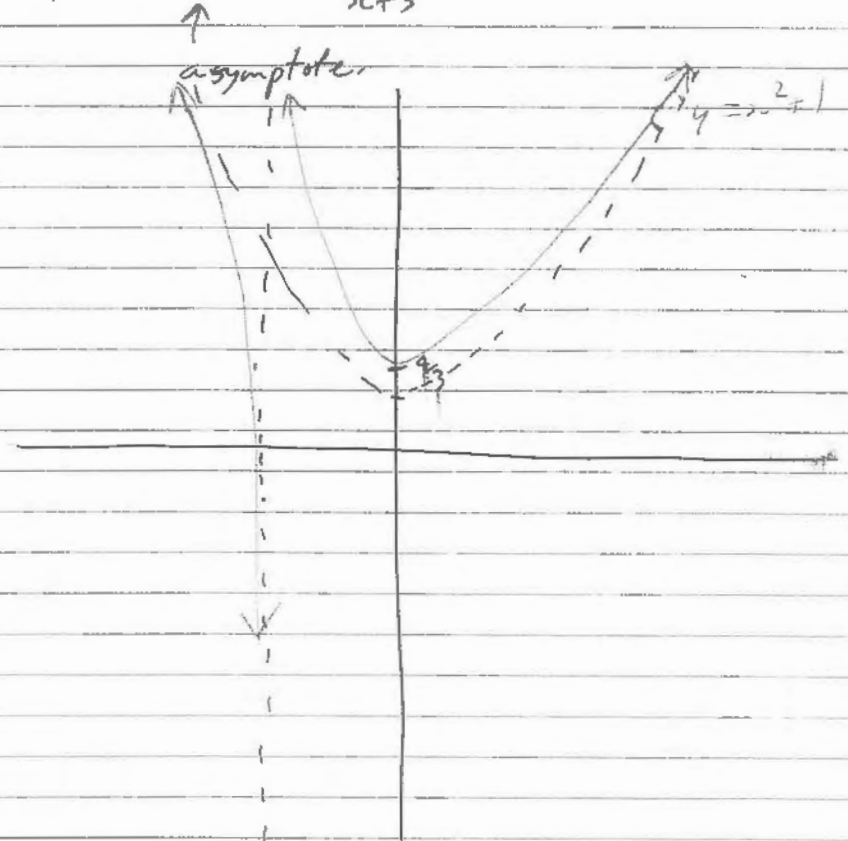
=

c) $y = \frac{x^3 + 3x^2 + x + 4}{x+3}$

$$\begin{array}{r} x^2 + 1 \\ x+3 \overline{) x^3 + 3x^2 + x + 4} \\ \underline{x^3 + 3x^2} \\ x + 4 \\ \underline{x + 3} \\ 1 \end{array}$$

$$\begin{array}{l} y_{\text{int}} \\ x=0 \\ y = 1 + \frac{1}{3} \\ = \frac{4}{3} \end{array}$$

$$y = x^2 + 1 + \frac{1}{x+3}$$



① let $z = x + iy$

$$3 |x + iy - (2 + 2i)| = |x + iy - (6 + 6i)|$$

$$\checkmark 3 \sqrt{(x-2)^2 + (y-2)^2} = \sqrt{(x-6)^2 + (y-6)^2}$$

$$3^2 (x^2 - 4x + 4 + y^2 - 4y + 4) = x^2 - 12x + 36 + y^2 - 12y + 36$$

$$9x^2 - 36x + 36 + 9y^2 - 36y + 36 = x^2 - 12x + 36 + y^2 - 12y + 36$$

$$\checkmark 8x^2 - 24x + 9y^2 - 24y = 0$$

$$x^2 - 3x + y^2 - 3y = 0$$

$$x^2 - 3x + \left(-\frac{3}{2}\right)^2 + y^2 - 3y + \left(-\frac{3}{2}\right)^2 = 2 \times \left(\frac{3}{2}\right)^2$$

$$(x - \frac{3}{2})^2 + (y - \frac{3}{2})^2 = \frac{9}{2}$$

$$\checkmark (x - \frac{3}{2})^2 + (y - \frac{3}{2})^2 = \left(\frac{3}{\sqrt{2}}\right)^2$$

$\checkmark$ sketch.

⑬ a) $I_n = \int x(x^5 + 1)^n dx$

$$v = (x^5 + 1)^n \quad u' = x$$

$$v' = 5x^4 n (x^5 + 1)^{n-1} \quad u = \frac{x^2}{2}$$

$$I_n = \frac{x^2}{2} (x^5 + 1)^n - \int \frac{x^2}{2} 5x^4 n (x^5 + 1)^{n-1} dx$$

$$= \frac{x^2}{2} (x^5 + 1)^n - \frac{5n}{2} \int x^6 (x^5 + 1)^{n-1} dx$$

$$= \frac{x^2}{2} (x^5 + 1)^n - \frac{5n}{2} \int x \cdot x^5 (x^5 + 1)^{n-1} dx$$

$$= \frac{x^2}{2} (x^5 + 1)^n - \frac{5n}{2} \int x [(x^5 + 1) - 1] (x^5 + 1)^{n-1} dx$$

$$= \frac{x^2}{2} (x^5 + 1)^n - \frac{5n}{2} \int x (x^5 + 1)^n - x (x^5 + 1)^{n-1} dx$$

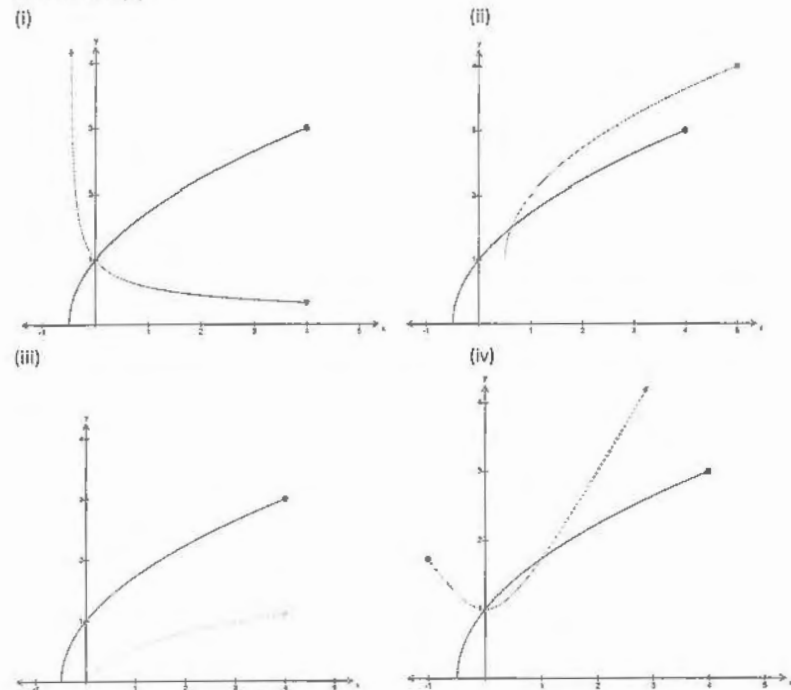
$$I_n = \frac{x^2}{2} (x^5 + 1)^n - \frac{5n}{2} I_n + \frac{5n}{2} I_{n-1}$$

$$\left(\frac{5n}{2} + 1\right) I_n = \frac{x^2}{2} (x^5 + 1)^n + \frac{5n}{2} I_{n-1}$$

$$\frac{5n+2}{2} I_n = \frac{x^2}{2} (x^5 + 1)^n + \frac{5n}{2} I_{n-1}$$

$$I_n = \frac{x^2 (x^5 + 1)^n}{5n+2} + \frac{5n}{5n+2} I_{n-1}$$

ANSWERS 13(b)



⑬ ② ⑪

$$I_{10} = \frac{x^2(x^5+1)^{10}}{52} + \frac{50}{52} I_9 \quad \checkmark$$

$$= \frac{x^2(x^5+1)^{10}}{52} + \frac{50}{52} \left[\frac{x^2(x^5+1)^9}{47} + \frac{45}{47} I_8 \right] \checkmark$$

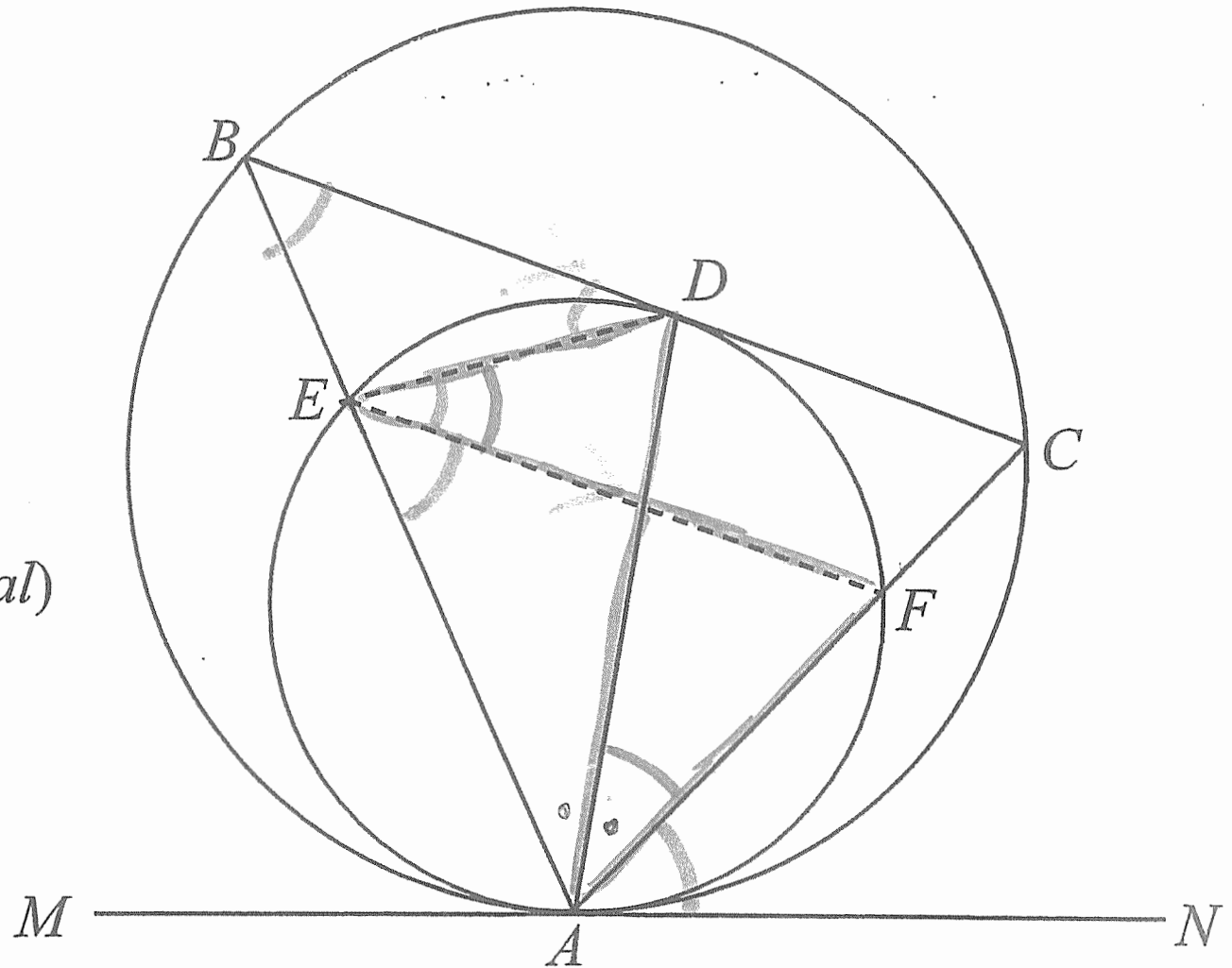
⑬ ⑥ see printed graphs

$\angle$ s DEF and BDE
 $\angle$ s DEF and DAF
 $\angle$ s BDE and DAE

1
 1
 1

and ED .
 Hence theorem in

$\angle$'s on transversal AB)
 $\angle$'s within parallel lines
 (equal)
 subtended by same arc DF at
 circumference of circle AEF are equal)
 Hence theorem in circle AEF)



- applies alternate segment theorem to deduce EF and BC are parallel
- deduces equality of angles DEF and BDE
- deduces equality of angles DEF and DAF
- deduces equality of angles BDE and DAE

Answer

Construct EF and ED .

Applying the alternate segment theorem in circles ABC and AEF :

$$\angle ABC = \angle NAC = \angle AEF$$

$\therefore EF \parallel BC$ (equal corresp. $\angle$'s on transversal AB)

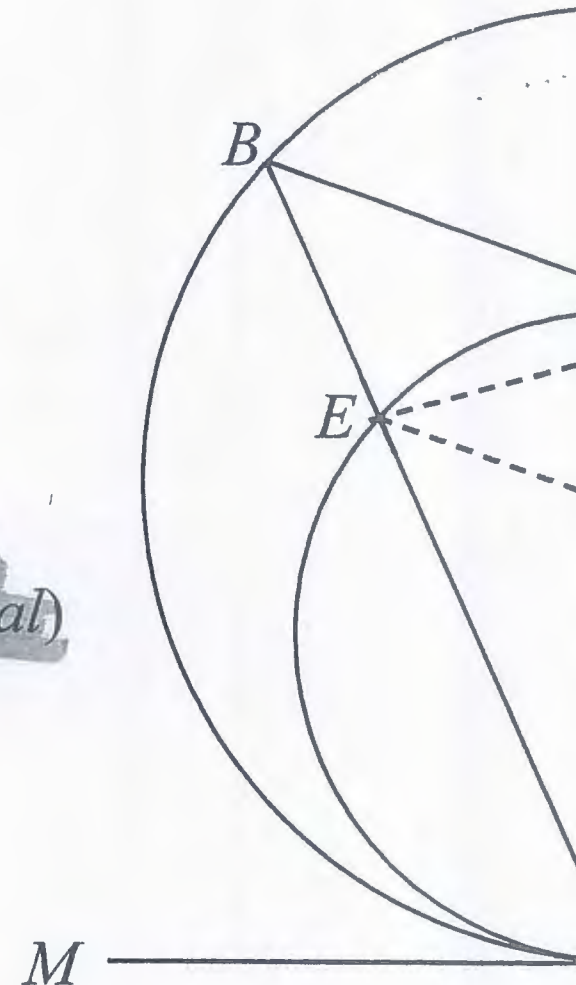
$\therefore \angle DEF = \angle BDE$ (Alt. $\angle$'s within parallel lines are equal)

② $\angle DEF = \angle DAF$ ($\angle$'s subtended by same arc DF at circumference of circle AEF are equal)

$\angle BDE = \angle DAE$ (Alt. segment theorem in circle AEF)

$$\therefore \angle DAF = \angle DAE \quad \angle DAF = \angle DEF = \angle BDE = \angle DAE$$

Hence AD bisects $\angle BAC$



(14) a

lower $< \int <$ upper.

$$1^2(p-1) < \int_1^p x^2 dx < p^2(p-1) \quad \checkmark$$

$$p-1 < \left[\frac{x^3}{3} \right]_1^p < p^3 - p^2 \quad \checkmark$$

$$p-1 < \frac{p^3}{3} - \frac{1}{3} < p^3 - p^2 \quad \checkmark$$

b For $n=1$

$$T_1 = 5 \times 1 - 2 = 3 \text{ as required.}$$

$$T_2 = 5 \times 2 - 2 = \text{" "}$$

fine for $n=1, 2$

assume true for $n=k-1$ $n=k-2$

$$\text{ie } T_k = 2T_{k-1}$$

$$T_{k-1} = 5(k-1) - 2 \quad T_{k-2} = 5(k-2) - 2.$$

for $n=k$ RTP $T_k = 5k - 2.$

$$T_k = 2T_{k-1} - T_{k-2}$$

$$= 2(5(k-1) - 2) - (5(k-2) - 2)$$

$$= 10k - 5k - 10 - 4 + 10 + 2$$

$$= 5k - 2$$

as required.

(14) c

$$P(x) = x^6 + ax^3 + bx^2$$

$$P(-1) = 1 - a + b = 0 \quad (1) \quad \checkmark$$

$$P'(x) = 6x^5 + 3ax^2 + 2bx$$

$$P'(-1) = -6 + 3a - 2b = 0 \quad (2) \quad \checkmark$$

from (1)

$$a = 1 + b$$

sub into (2)

$$-6 + 3(1+b) - 2b = 0$$

$$-3 + 3b - 2b = 0$$

$$b = 3$$

$$a = 1 + 3 \quad \checkmark$$

(14) d Product = e_a

$$k = \alpha \frac{1}{\alpha} \beta \frac{1}{\beta}$$

$$= 1 \quad \checkmark$$

(ii) ~~α for example~~

~~satisfys~~

sub in $\alpha.$

$$\alpha^4 + b\alpha^3 + c\alpha^2 + d\alpha + 1 = 0$$

sub in $\frac{1}{\alpha}$

$$(2) \quad x^3 + 3x^2 + 2x + 1 = 0 \quad \alpha^2, \beta^2, \gamma^2$$

$$\text{let } X = x^2 \\ \sqrt{X} = x \\ x = X^{1/2}$$

$$(X^{1/2})^3 + 3(X^{1/2})^2 + 2(X^{1/2}) + 1 = 0 \quad \checkmark$$

$$X^{1/2}(X+2) = -(1+3X) \quad \checkmark$$

square

$$X(X^2 + 4X + 4) = 1 + 6X + 9X^2$$

$$X^3 + 4X^2 + 4X = 1 + 6X + 9X^2$$

$$X^3 - 5X^2 + 10X - 1 = 0$$

$$X^3 - 5X^2 - 2X - 1 = 0 \quad \checkmark$$

$$(13) (a) (i) \quad x = \frac{at}{2} \quad y = \frac{a}{t}$$

$$\frac{2x}{a} = t \quad t = \frac{a}{y}$$

$$\frac{2x}{a} = \frac{a}{y}$$

$$xy = \frac{a^2}{2} \quad \checkmark$$

$$(ii) \quad \frac{dx}{dt} = \frac{a}{2} \quad y = at^{-1} \\ \frac{dy}{dt} = -at^{-2} = -\frac{a}{t^2}$$

$$\frac{dy}{dx} = -\frac{a}{t^2} \div \frac{a}{2}$$

$$= -\frac{a}{t^2} \times \frac{2}{a}$$

$$= -\frac{2}{t^2} \quad \checkmark$$

$$\text{eqn } y - \frac{a}{t} = -\frac{2}{t^2} \left(x - \frac{at}{2} \right)$$

$$t^2 y - at = -2x + at \quad \checkmark$$

$$2x + t^2 y = 2at$$

(iv)

$$m_{\text{tangent}} + m_{\perp} = -1$$

$$-\frac{2}{t^2} + m_{\perp} = -1$$

$$m_{\perp} = \frac{t^2}{2}$$

$$y - a = \frac{t^2}{2}(x - a)$$

$$2y - 2a = t^2x - at^2$$

$$at^2 - 2a = t^2x - 2y$$

$$t^2x - 2y = at^2 - 2a$$

(iv) T lies on the 2 lines from (ii) & (iii)

$$\begin{cases} 2x + t^2y = 2at & (1) \\ t^2x - 2y = at^2 - 2a & (2) \end{cases}$$

Square (1) & (2)

$$(1)^2 \quad (2x)^2 + 2 \times 2x \times t^2y + t^4y^2 = 4a^2t^2$$

$$4x^2 + 4xt^2y + t^4y^2 = 4a^2t^2 \quad (3)$$

$$(2)^2 \quad t^4x^2 - 4t^2xy + 4y^2 = a^2t^4 - 4a^2t^2 + 4a^2 \quad (4)$$

$$(3) + (4) \quad 4x^2 + t^4x^2 + 4y^2 + t^4y^2 = 4a^2 + 4a^2t^2$$

$$x^2(4+t^4) + y^2(4+t^4) = a^2(4+t^4)$$

$$\div 4+t^4$$

$$x^2 + y^2 = a^2$$

$$b) (1) \left(\cos \frac{x}{2} + i \sin \frac{x}{2} \right)^2$$

$$= \cos^2 \frac{x}{2} + i \sin 2 \frac{x}{2}$$

$$= \cos x + i \sin x$$

Also

$$\left(\cos \frac{x}{2} + i \sin \frac{x}{2} \right)^2 = \cos^2 \frac{x}{2} + 2i \sin \frac{x}{2} \cos \frac{x}{2} - \sin^2 \frac{x}{2} \quad \checkmark$$

$$\text{Real} \quad \cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$$

$$\text{Im} \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} \quad \checkmark$$

$$(1) \quad t = \tan \frac{x}{2} \quad \tan x = \frac{\sin x}{\cos x}$$

$$= \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}$$

$$(1) \quad \tan x = \frac{\sin x}{\cos x}$$

$$= \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}$$

$$\div \cos^2 \frac{x}{2} = \frac{2 \tan \frac{x}{2}}{1 - \tan^2 \frac{x}{2}}$$

$$= \frac{2t}{1-t^2} \quad \checkmark$$

Q.1 $z^7 = 1$

$$z = 1, \text{cis } \frac{2\pi n}{7}$$

$n = \pm 1, \pm 2, \pm 3$

OR

$$z = \text{cis } \frac{2\pi n}{7} \quad n = 0, \pm 1, \pm 2, \pm 3$$

(ii) $\frac{1}{2} (\cos n\theta + i \sin n\theta)$

$$= \frac{1}{2} (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta) \quad \checkmark$$

$$= \frac{1}{2} \times 2 \cos n\theta$$

$$= \cos n\theta$$

$$\frac{1}{2} + \cos \alpha + \cos 2\alpha + \cos 3\alpha$$

(iii) $\frac{1}{2} (1 + w + w^{-1} + w^2 + w^{-2} + w^3 + w^{-3}) = 0$

$$\times \frac{1}{2} \times 2$$

$$(1 + w + w^{-1} + w^2 + w^{-2} + w^3 + w^{-3}) = 0$$

$$w^{-3} \times w^{-2} \times w^{-1} \times 1 \times w \times w^2 \times w^3 = 0$$

$$+ w^3 \quad 1 \times w \times w^2 \times w^3 \times w^4 \times w^5 \times w^6 = 0$$

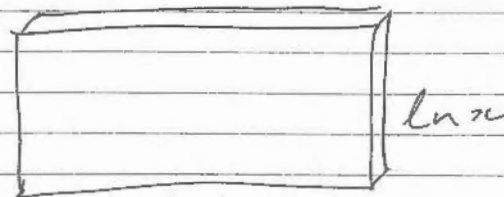
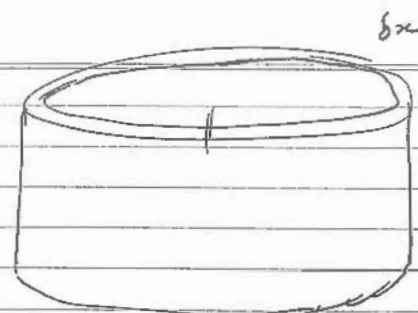
$$\frac{w^7 - 1}{w - 1} = 0$$

$$w^7 = 1$$

(iv) $\alpha = \frac{2\pi n}{7} \quad n = \pm 1, \pm 2, \pm 3$

NOT $n = 0$

Q.2



$$2\pi \left(\frac{\ln x + 1}{2} \right) 2\pi (x + \frac{1}{2})$$

$$\delta V = 2\pi (x + 1) \ln x \delta x \quad \checkmark$$

$$V = 2\pi \int_1^B (x \ln x + \ln x) dx$$

$$u = \ln x \quad u' = \frac{1}{x}$$

$$v' = \frac{1}{x} \quad u = \frac{x^2}{2}$$

$$v = \ln x$$

$$u' = 1$$

$$v' = \frac{1}{x}$$

$$u = x$$

$$= 2\pi \left[\frac{x^2}{2} \ln x + x \ln x \right]_1^B - \int_1^B \frac{x^2}{2} + 1 dx$$

$$= 2\pi \left(\left[\frac{e^2}{2} + e - 0 \right] - \left[\frac{e^2}{4} + e - \frac{1}{4} - 1 \right] \right)$$

$$= 2\pi \left[\frac{e^2}{2} + e - \left(\frac{e^2}{4} + e - \frac{1}{4} - 1 \right) \right]$$

$$= 2\pi \left(\frac{Be^2}{4} + \frac{5}{4} \right) \quad \checkmark$$

$$\textcircled{1} \Sigma F = 100 - kv^2$$

$$F = ma$$

$$100 - kv^2 = 20 \ddot{x}$$

$$5 - \frac{kv^2}{20} = \ddot{x}$$

$$\textcircled{II} \ddot{x} = 5 - \frac{kv^2}{20}$$

$$v \frac{dv}{dx} = 5 - \frac{kv^2}{20}$$

$$\frac{dv}{dx} = \frac{5}{v} - \frac{kv}{20}$$

$$\frac{dv}{dx} = \frac{100 - kv^2}{20v}$$

$$\frac{dx}{dv} = \frac{20v}{100 - kv^2}$$

$$x = \frac{10}{k} \int \frac{-2kv}{100 - kv^2} dv$$

$$= -\frac{10}{k} \ln(100 - kv^2) + C$$

$$\text{sub } x=0 \quad v=0$$

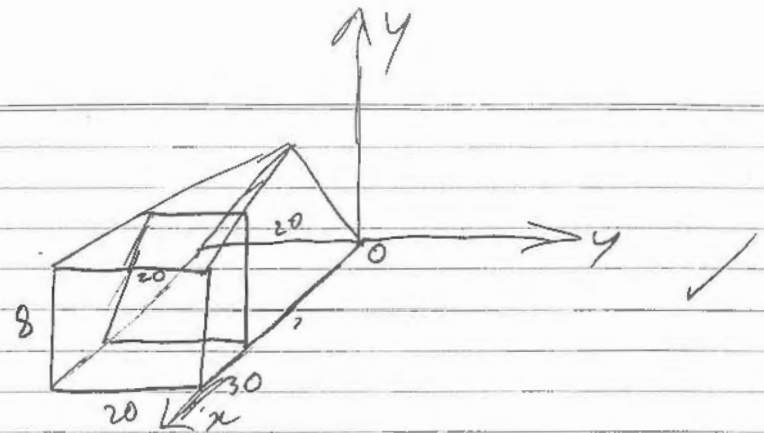
$$C = \frac{10}{k} \ln 100$$

$$x = \frac{10}{k} (\ln 100 - \ln(100 - kv^2))$$

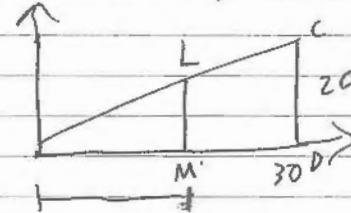
$$= \frac{10}{k} \ln \left(\frac{100}{100 - kv^2} \right)$$

(16c)

(I)



(II) JK is clearly constant 20cm ✓



$$m = \frac{20}{30} = \frac{2}{3}$$

equation 2

$$y = \frac{2}{3}x$$

$$ML = \frac{2}{3}x$$

$$\textcircled{III} \text{Area} = \frac{1}{2} \left(\frac{2}{3}x + 20 \right) \times 8$$

$$\textcircled{IV} V = \int_0^{30} \frac{1}{2} \left(\frac{2}{3}x + 20 \right) \times 8 dx$$

$$= 4 \int_0^{30} \left(\frac{2}{3}x + 20 \right) dx$$

$$= 4 \left[\frac{2x^2}{3} + 20x \right]_0^{30}$$

$$= 4 (300 + 600)$$

$$= 3600 \text{ m}^3$$